



$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + 5x} - \sqrt{x^2 - 3x})$$

$$\begin{pmatrix} 0 \\ 0 \\ \text{L'H. R.} \end{pmatrix} ; \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \left(\frac{\cancel{x^2} + 5x - (\cancel{x^2} + 3x)}{\sqrt{x^2 + 5x} + \sqrt{x^2 - 3x}} \right) =$$

$$= \lim_{x \rightarrow \infty} \frac{8x}{\sqrt{x^2 + 5x} + \sqrt{x^2 - 3x}} =$$

$$\lim_{x \rightarrow \infty} \frac{8\cancel{x}}{\cancel{x}\sqrt{1 + \frac{5}{x}} + \cancel{x}\sqrt{1 - \frac{3}{x}}} = \frac{8}{\sqrt{1 + \frac{5}{x}} + \sqrt{1 - \frac{3}{x}}}$$

$$\lim_{x \rightarrow \infty} \frac{8}{\sqrt{1 + \frac{5}{x}} + \sqrt{1 - \frac{3}{x}}} = \frac{8}{\lim_{x \rightarrow \infty} \sqrt{1 + \frac{5}{x}} + \lim_{x \rightarrow \infty} \sqrt{1 - \frac{3}{x}}}$$

$$= \frac{8}{1 + 1} = \frac{8}{2} = 4$$

$$\left((1+3x)^{1/2} \right)' = \frac{1}{2} (1+3x)^{1/2-1}$$

④ $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+3x} - 1}{x} =$

$= \lim_{x \rightarrow 0} \frac{\frac{1}{3} \cdot 1}{(1+3x)^{2/3}} = \lim_{x \rightarrow 0} \frac{1}{(1+3x)^{2/3}} =$

$= 1$

$$\begin{aligned} & a^3 - b^3 \\ & (a-b)(a^2 + ba + b^2) \end{aligned}$$

$y = \frac{2x+1}{x-3}$

$k = 2$

$b = 4$

$\lim_{x \rightarrow \infty} \frac{2x+1 - 2x(x-3)}{x-3} =$

$= \lim_{x \rightarrow \infty} \frac{2x+1 - 2x^2 + 6x}{x-3} =$

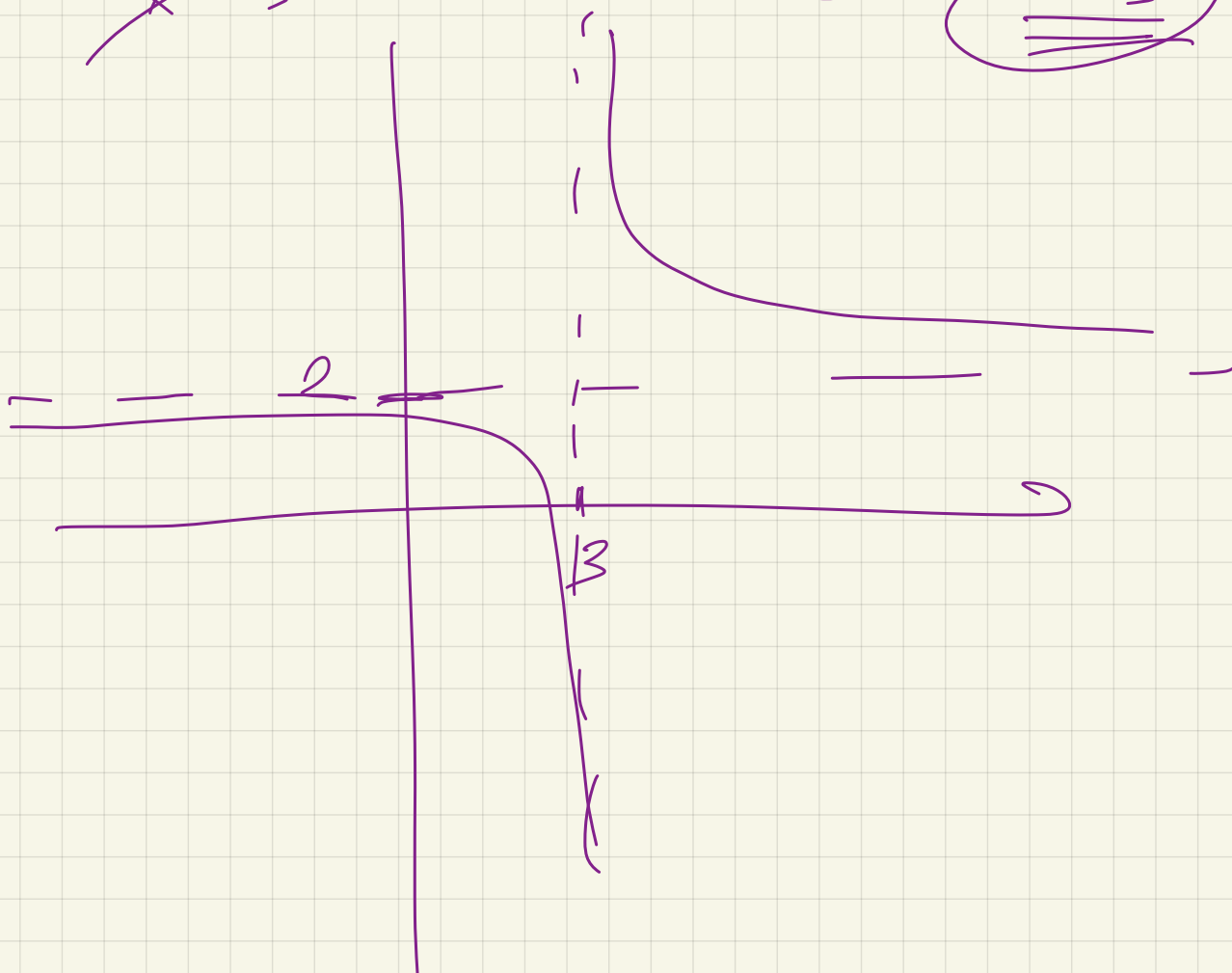
$= \lim_{x \rightarrow \infty} \frac{-2x^2 + 8x + 1}{x-3} = \lim_{x \rightarrow \infty} \frac{-2x + 8}{1} = \infty$

$$\lim_{x \rightarrow \infty} \frac{2x+1}{x^2-3x} = \lim_{x \rightarrow \infty} \frac{2}{2x-3} = 0$$

$$\left(\frac{2x+1}{x-3} \right) \frac{1}{x} \quad \frac{2x+1}{x-3} : x$$

$$y = \frac{2x+1}{x-3} = \frac{(2x-6)+6+1}{x-3} =$$

$$\frac{2(x-3)}{x-3} + \frac{6+1}{x-3} = 2 + \frac{7}{x-3}$$



$$y = \frac{x^2 + 1}{x + 2}$$

$$\underline{x = -2}$$

$$K = \lim_{x \rightarrow \infty} \frac{x^2 + 1}{x + 2} \cdot \frac{1}{x} = \lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^2 + 2x} =$$

$$\lim_{x \rightarrow \infty} \frac{2x + 1}{2x + 2} = \lim_{x \rightarrow \infty} \frac{2}{2} = 1$$

$$K = 1$$

$$b = \lim_{x \rightarrow \infty} \frac{x^2 + 1}{x + 2} - x = \lim_{x \rightarrow \infty} \frac{x^2 + 1 - x(x + 2)}{x}$$

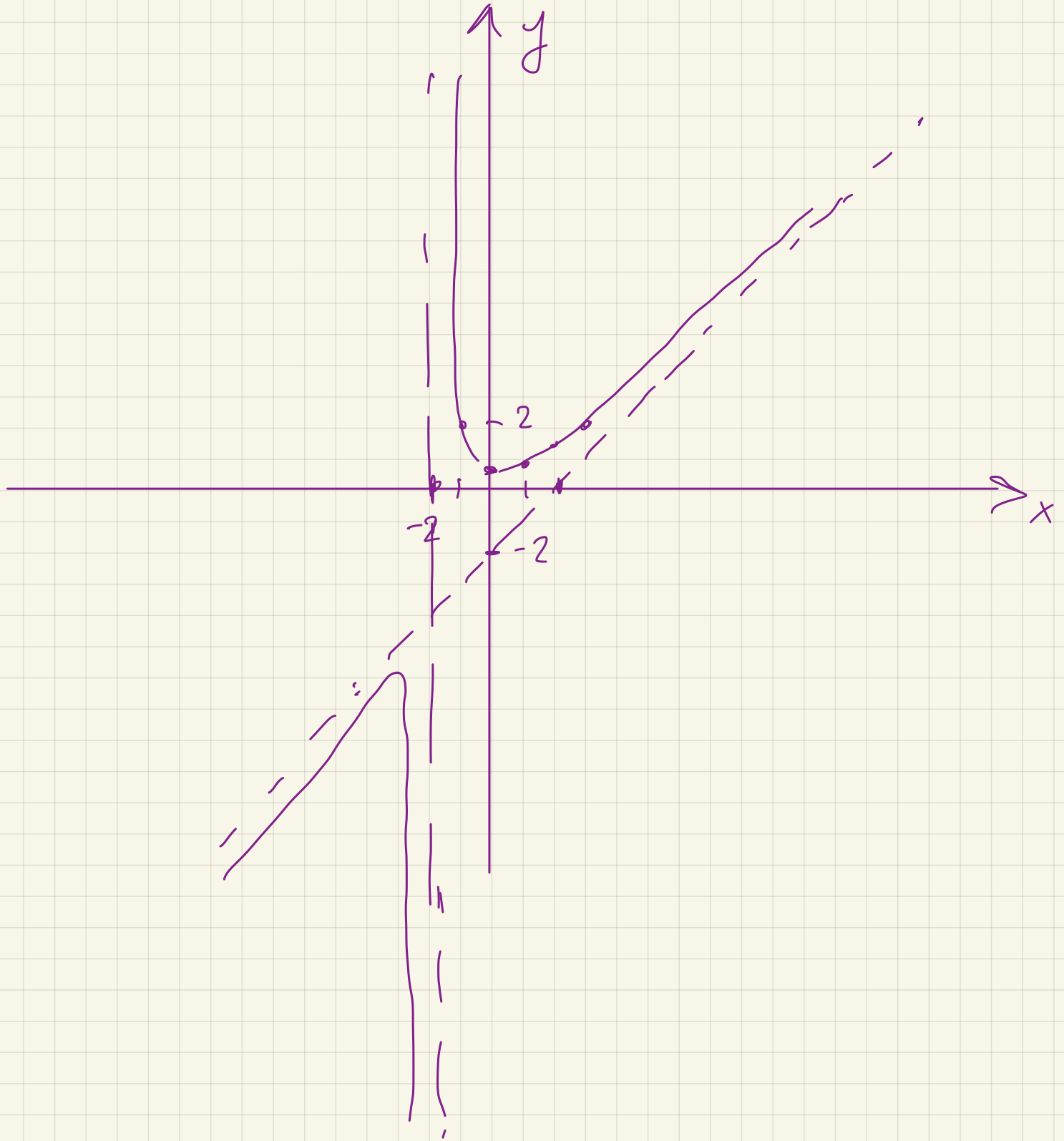
$$= \lim_{x \rightarrow \infty} \frac{\cancel{x^2} + 1 - \cancel{x^2} - 2x}{x + 2} =$$

$$= \lim_{x \rightarrow \infty} \frac{-2x + 1}{x + 2} = \lim_{x \rightarrow \infty} -\frac{2}{1} = \textcircled{-2}$$

$$y = x - 2 \quad \text{— Hakenwert}$$

$$y = \frac{x^2 + 1}{x + 2}$$

y	1/2	2	2/3	5/4	12
x	0	-1	1	2	3



$$y = \frac{x^2 - 2x + 3}{x - 1}$$

$$k = 1$$

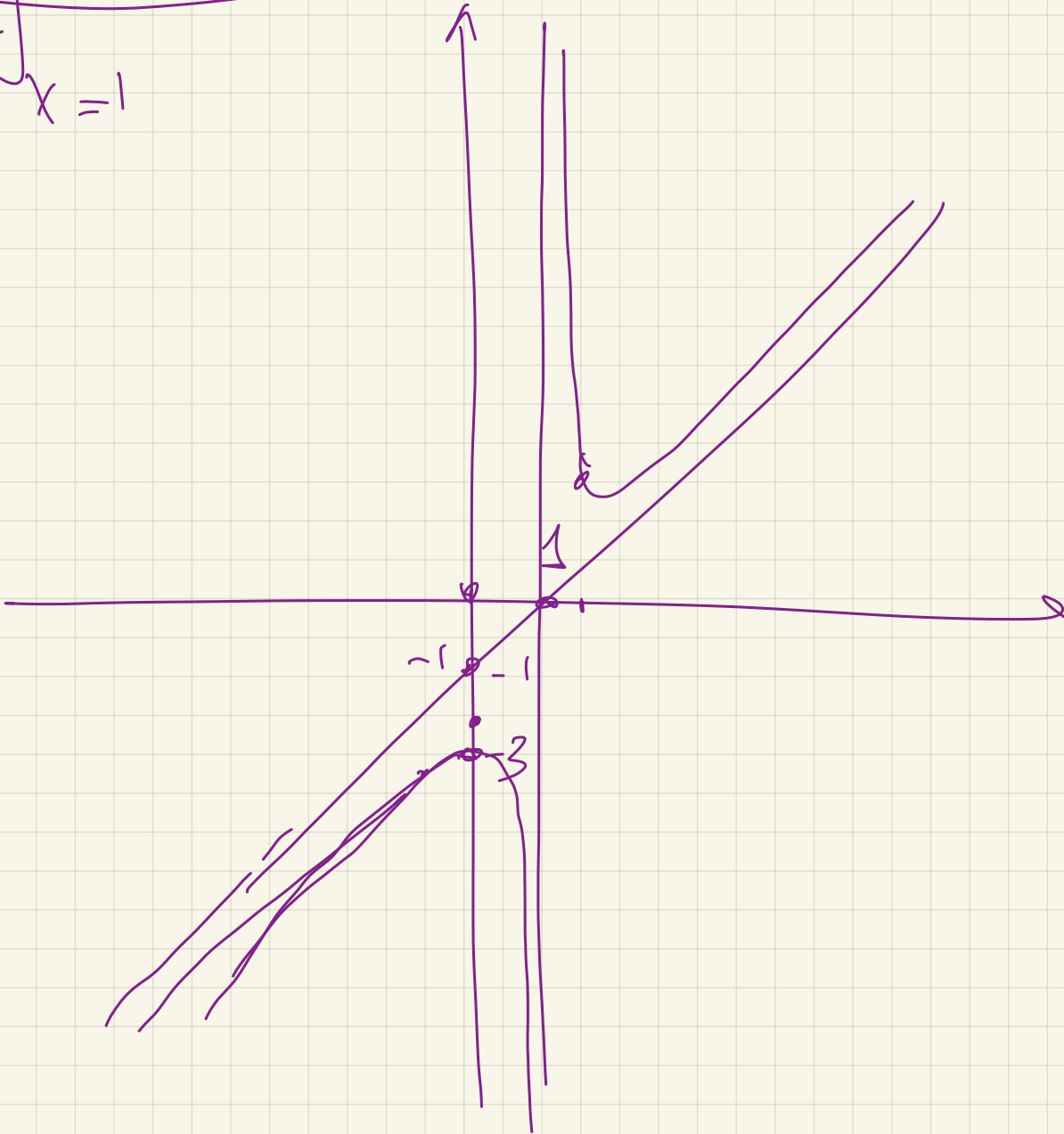
$$\lim_{x \rightarrow \infty} \frac{x^2 - 2x + 3 - x^2 + x}{x - 1} = \lim_{x \rightarrow \infty} \frac{-x + 3}{x - 1}$$

$$\lim_{x \rightarrow \infty} -\frac{1}{y} = -1$$

x	0	$-\frac{11}{3}$
y	-3	-2

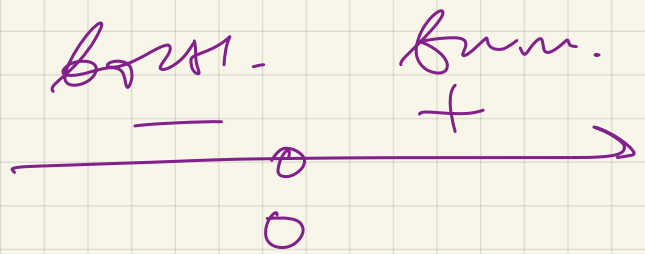
$y = x - 1$ ← remu. acc.

$x = 1$



$$y'' = \left(\frac{2}{x^3} \right) \neq 0$$

$x \neq 0$



$$y = \frac{x}{x^2+1} \Rightarrow y' = \frac{1(x^2+1) - 2x \cdot x}{(x^2+1)^2}$$

$$= \frac{-x^2+1}{(x^2+1)^2}$$

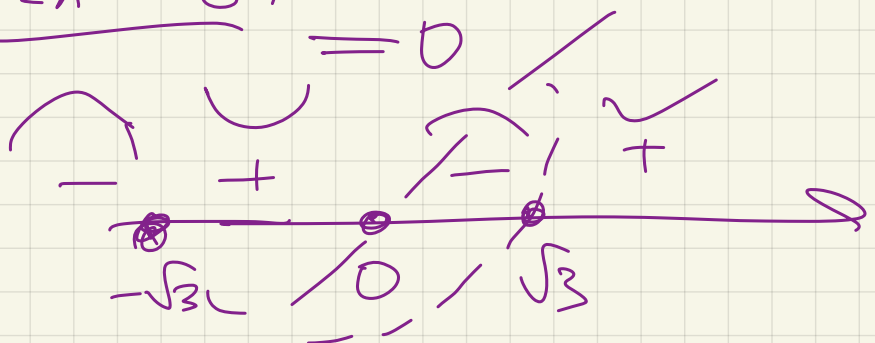
$$y'' = \frac{-2x(x^2+1)^2 - 2(x^2+1) \cdot 2x(-x^2+1)}{(x^2+1)^4}$$

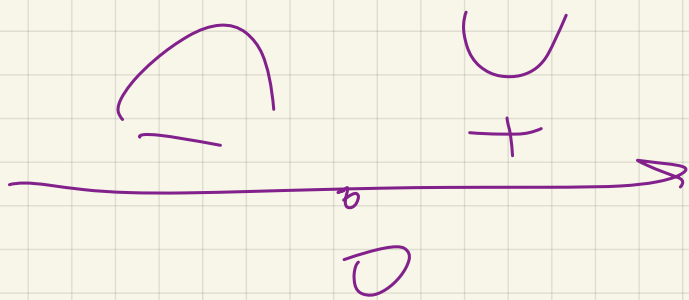
$$\frac{x(x^2+1)(-2(x^2+1) - 4(1-x^2))}{(x^2+1)^4}$$

$$x^2 = 3$$

$$\frac{x(x^2+1)(2x^2-6)}{(x^2+1)^3} = 0$$

$$x^2+1 \neq 0$$





Однородные ϕ -ин?

$$f(kx; ky) = k^t f(x, y)$$

$$f(x, y) = x^2 + 2xy + y^2$$

$$x \Rightarrow kx$$

$$y \Rightarrow ky$$

$$(kx)^2 + 2kx \cdot ky + (ky)^2 =$$

$$k^2 x^2 + 2k^2 xy + k^2 y^2 =$$

$$= \underline{\underline{k^2 (x^2 + 2xy + y^2)}}$$

однородные

$$f(x, y) = x^3 + 2xy$$

$$(kx)^3 + 2kx \cdot ky =$$

$$k^3 x^3 + 2k^2 xy \quad \text{неоднородно}$$

$$f(x,y) = \sqrt{x^2 + y^2}$$

(+)

$$f(-x,y) = -12x^8 + 6x^4y^4 + x^2y(x^5 - 3xy^4) + 2022y^8$$

$\begin{matrix} & & & & k^8 \\ & & & & \nearrow \\ & & & & x^2y(x^5 - 3xy^4) \\ & & & & \downarrow \\ & & & & k^8 \end{matrix}$

8 power